

DIGITAL IMAGE PROCESSING

LECTURE # 6

IMAGE ENHANCEMENT IN SPATIAL DOMAIN-III

Topics to Cover

- Sharpening Spatial Filtering
- Edge Detection
- Derivatives
 - ❑ 1st Order Derivative
 - ❑ 2nd Order Derivative
- Laplacian Operator
- Unsharp Masking
- High Boost Filtering

Spatial Filtering for Sharpening

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Background: To highlight fine detail in an image or to enhance blurred detail

Applications: Medical imaging, industrial inspection etc.

Foundation (Blurring vs Sharpening):

- Blurring/smoothing is performed by spatial averaging (equivalent to integration)
- Sharpening is performed by noting only the gray level changes in the image that is the differentiation



Edge Detection

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What is an Edge?

Edge is a change but every change is not an edge

Edge is a noticeable or abrupt change (equivalent to High pass filter in frequency domain)

E.g.

5	7
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 2 is not a noticeable change in the range of (0 to 255)

We have to define a threshold if the change is more than a specified threshold then we will define it as an edge point.

Here gradual change exists you cannot pinpoint where the edge exists so the change must be abrupt

50	55	60	65	70
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For each pixel we have to look in horizontal, vertical and diagonal direction

dx/ds -> for horizontal direction

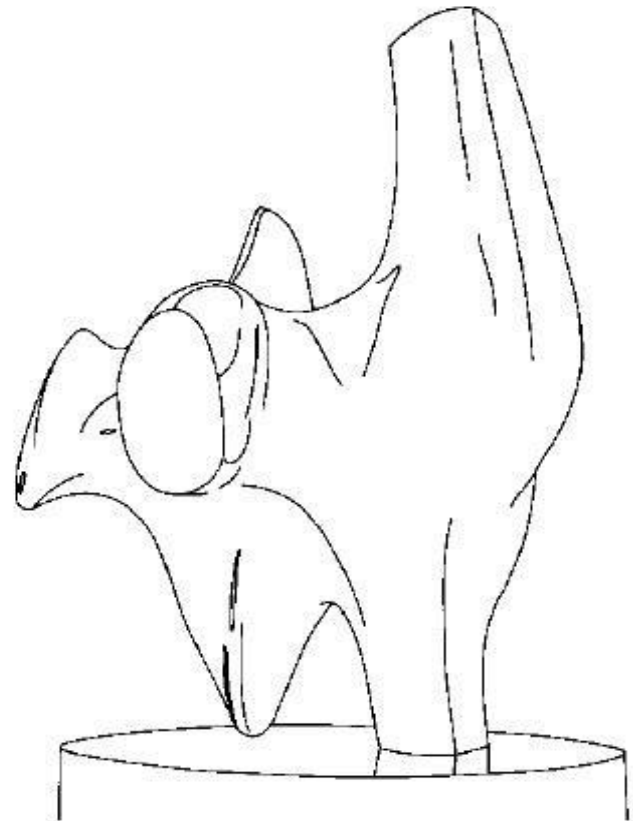
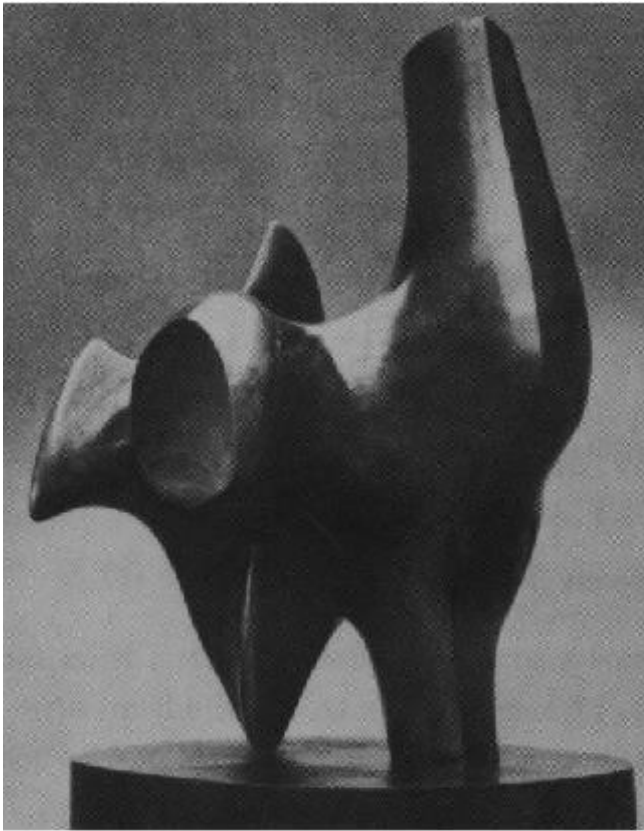
dy/ds -> for vertical direction

dd/ds -> for diagonal direction

50	15	35
40	20	30
225	195	230

Edge Detection

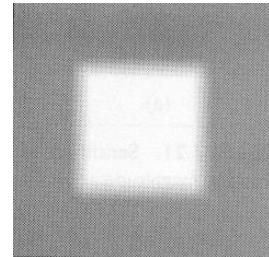
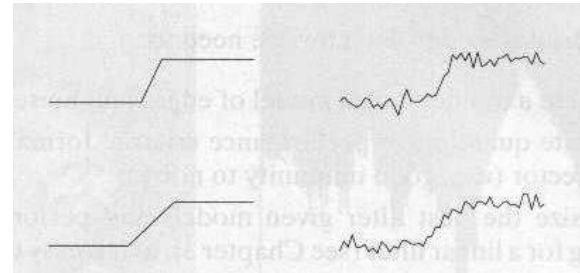
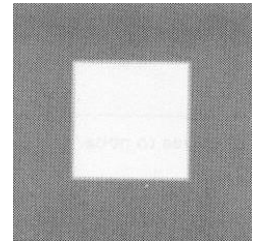
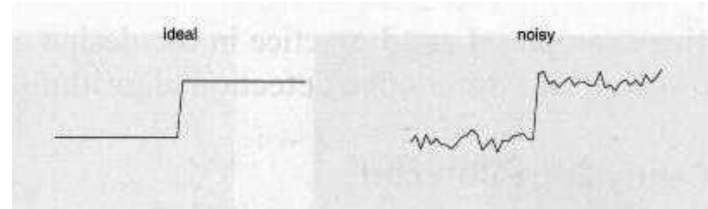
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Edge Types

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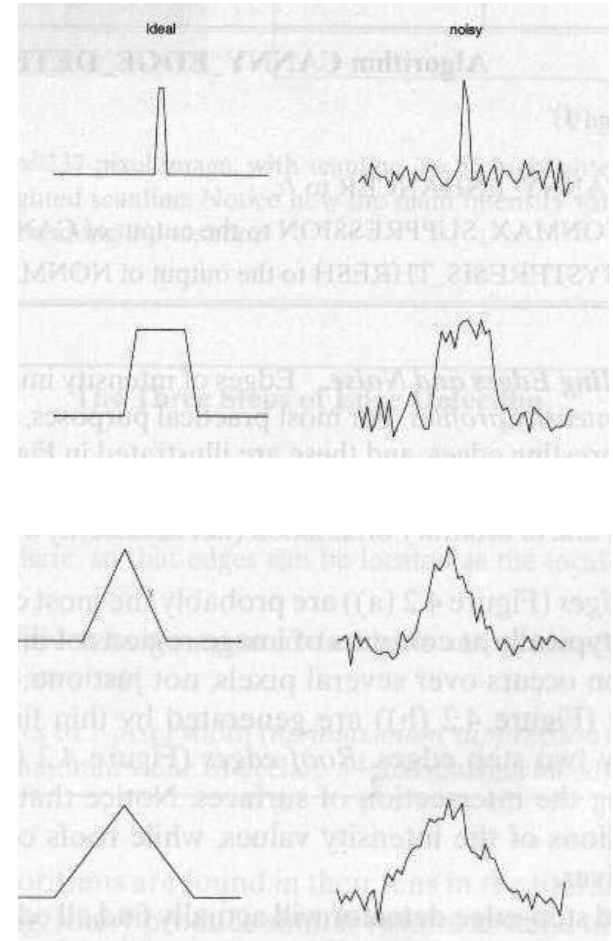
- **Step edge:** the image intensity abruptly changes from one value on one side of the discontinuity to a different value on the opposite side.
- **Ramp edge:** a step edge where the intensity change is not instantaneous but occurs over a finite distance.



Edge Types

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- ❑ Ridge edge: the image intensity abruptly changes value but then returns to the starting value within some short distance (i.e., usually generated by lines).
- ❑ Roof edge: a ridge edge where the intensity change is not instantaneous but occur over a finite distance.



Spatial Filtering for Sharpening

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Mathematical Basis of Filtering for Image Sharpening

- First-order derivatives [Gradient]
- Second-order derivatives

Derivatives

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First Order Derivative

- A basic definition of the first-order derivative of a one-dimensional function $f(x)$ is the difference

$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

Second Order Derivative

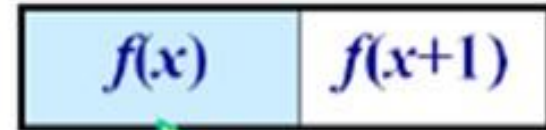
- Similarly, we define the second-order derivative of a one-dimensional function $f(x)$ is the difference

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1) + f(x - 1) - 2f(x)$$

1st & 2nd Order Derivatives

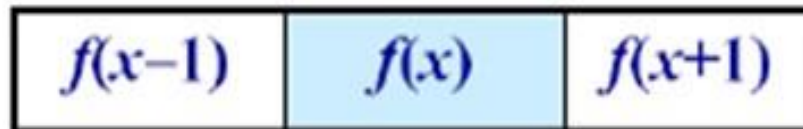
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$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$



Position for the
output pixel

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$



1st & 2nd Order Derivatives

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First Order Derivative

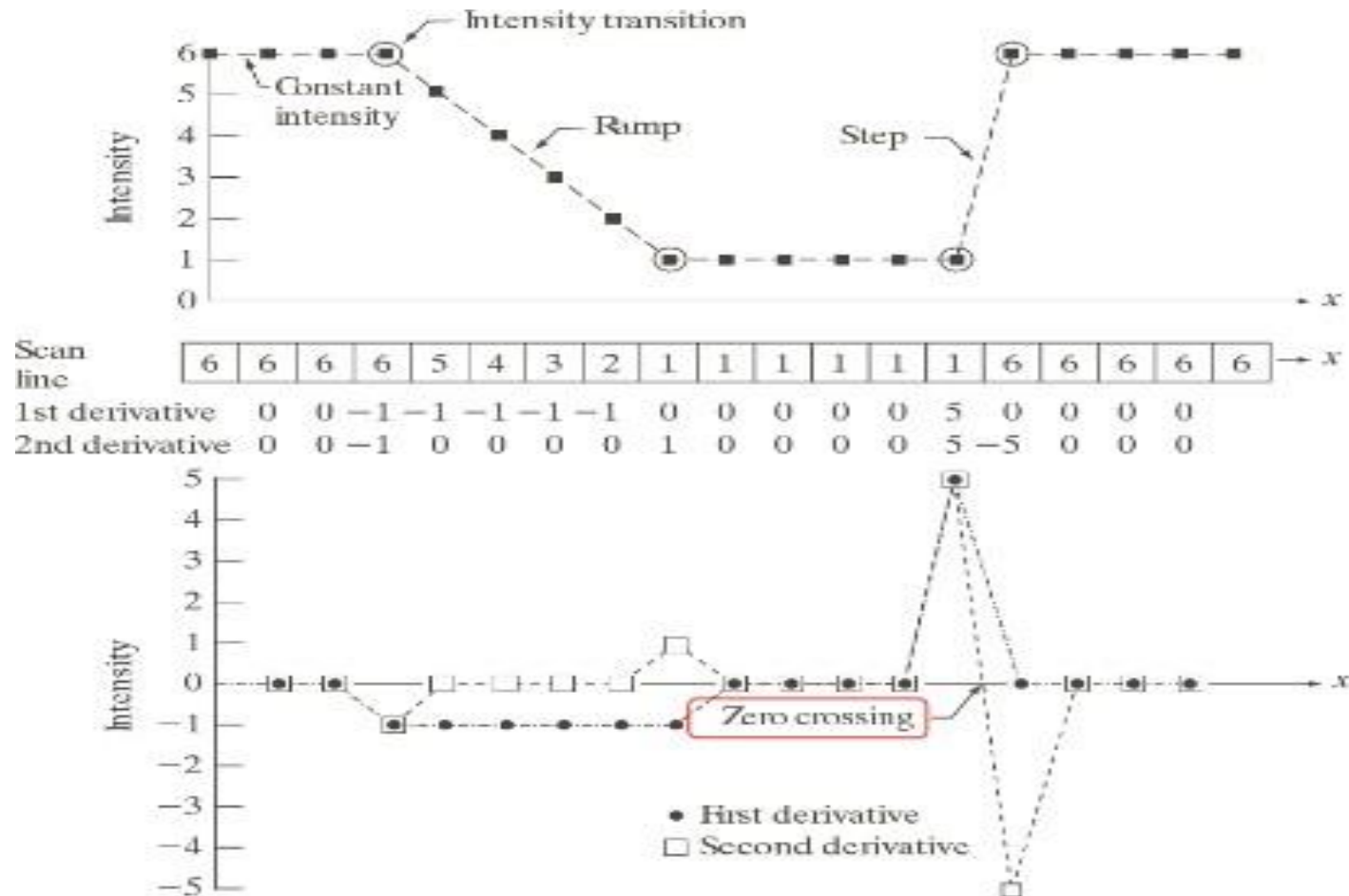
- ❑ Zero in area of constant gray levels
- ❑ Non zero along the ramps
- ❑ Non zero at the start of the gray level step or ramp

Second Order Derivative

- ❑ Zero in flat areas
- ❑ Zero along the ramps of constant slope
- ❑ Non zero at the start and end of the gray level step or ramp

Example for Discrete Derivatives

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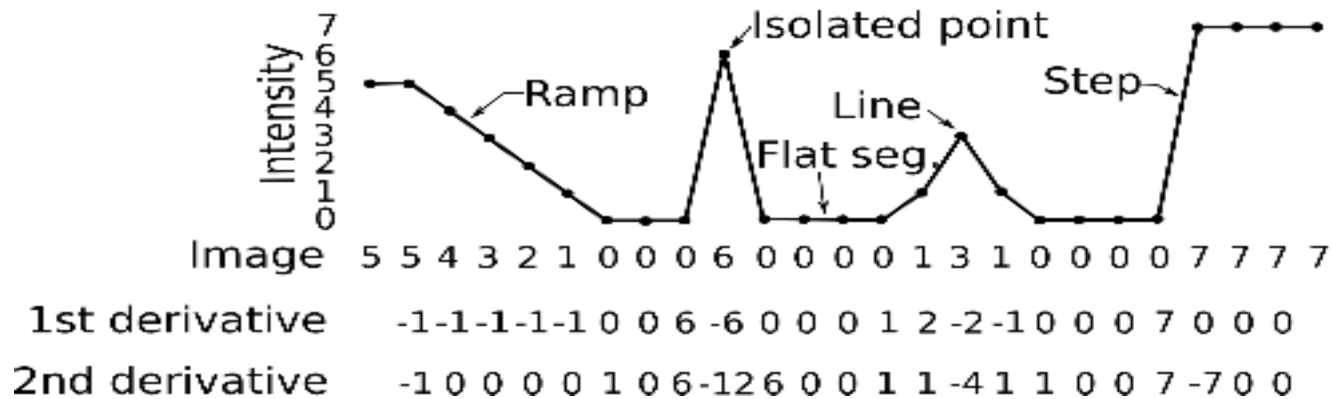
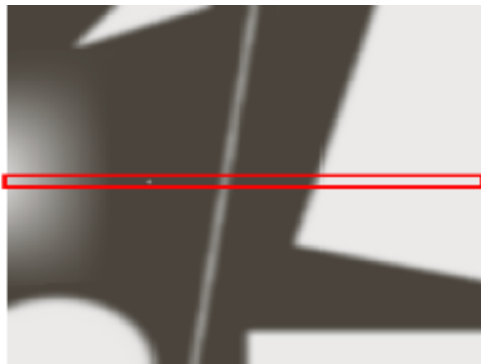


a
b
c

FIGURE 3.36 Illustration of the first and second derivatives of a 1-D digital function representing a section of a horizontal intensity profile from an image. In (a) and (c) data points are joined by dashed lines as a visualization aid.

Example for Discrete Derivatives

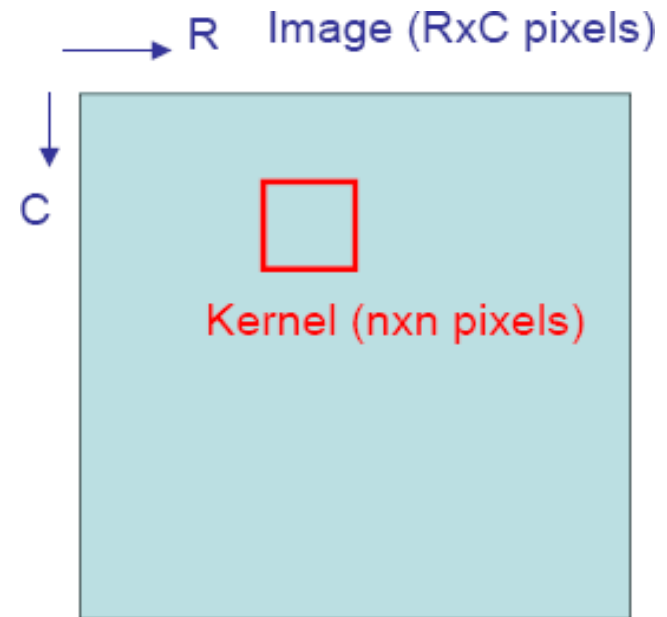
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Steps to Apply Edge Detector Kernel

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1. Suppose a $n \times n$ (n is an odd number) kernel is used and the centre position of the kernel is denoted by k_c .
2. Slide the kernel across the image, one pixel at a time.
3. Compute the new value of each pixel, which is under k_c , and repeat for all pixels in the image.
4. Note that the process is not in-place (new values are stored in a new image of same dimension).
5. Computation:
 - For each pixel inside the kernel, multiply the pixel value with the kernel value.
 - Take the sum of the products.
6. (a) The border pixel can be duplicated, or
(b) take the mirror values of the border pixels, to fill up the kernel.



Laplacian Mask: 2nd Order Derivative

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$$f = f(x, y)$$

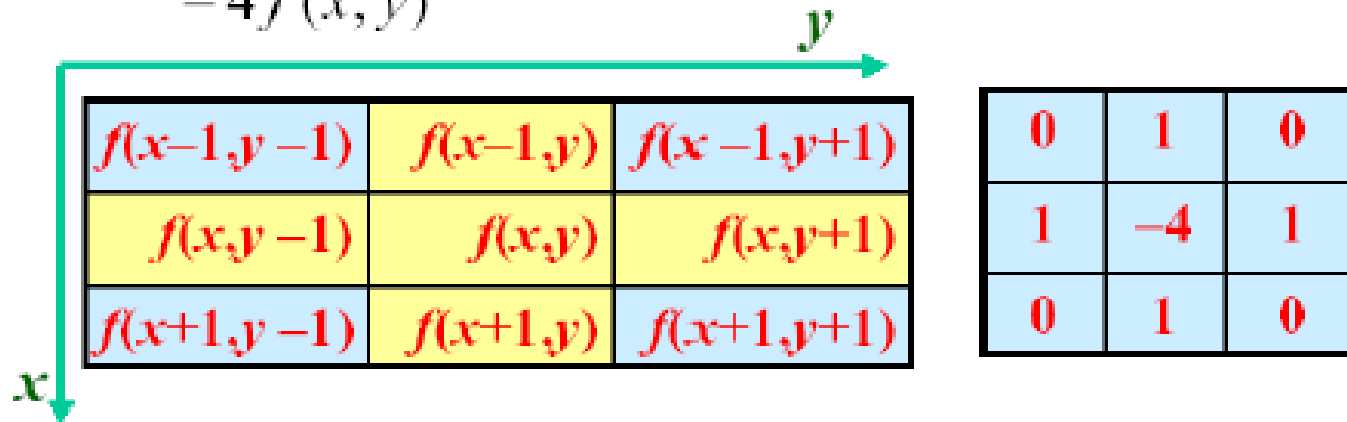
Laplacian operator

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$



Laplacian Mask: 2nd Order Derivative

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0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

a	b
c	d

FIGURE 3.39

(a) Filter mask used to implement the digital Laplacian, as defined in Eq. (3.7-4).
(b) Mask used to implement an extension of this equation that includes the diagonal neighbors. (c) and (d) Two other implementations of the Laplacian.

Laplacian for Image Enhancement

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- Image background is removed by Laplacian filtering.
- Background can be recovered simply by adding original image to Laplacian output

To obtain the enhanced image $\hat{p}$

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y), w_5 < 0 \\ f(x, y) + \nabla^2 f(x, y), w_5 > 0 \end{cases}$$

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

In this way, background tonality can be perfectly preserved while details are enhanced

Laplacian for Image Enhancement

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a b
c d

FIGURE 3.40
(a) Image of the North Pole of the moon.
(b) Laplacian-filtered image.
(c) Laplacian image scaled for display purposes.
(d) Image enhanced by using Eq. (3.7-5).
(Original image courtesy of NASA.)

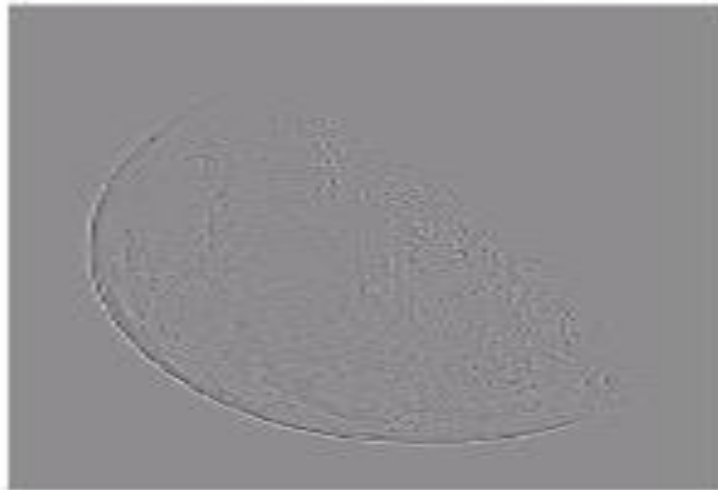


Image Sharpening Based on Un-sharp Masking

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Un-sharp masking

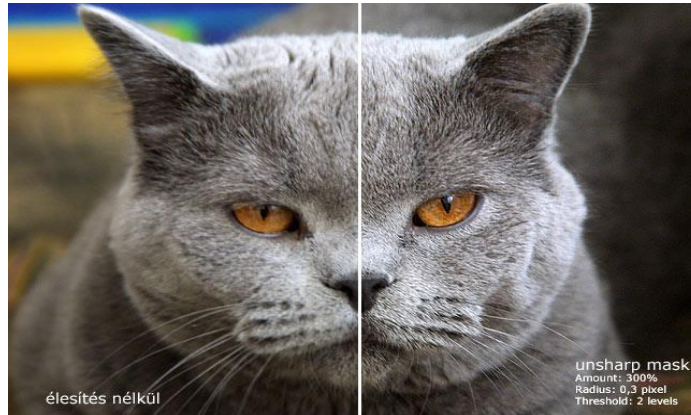
- Sharpen images consists of subtracting an unsharp (smoothed) version of an image from the original image

Subtract the blurred image from the original image

$$f_s(x, y) = f(x, y) - \bar{f}(x, y)$$

$f_s(x, y) \Rightarrow$ sharpened image

$\bar{f}(x, y) \Rightarrow$ blurred image of $f(x, y)$



High Boost Filtering

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- A slight further generalization of un-sharp masking is called High Boost filtering
- A high boost filtered image, f_{hb} , is defined at any point (x, y) as

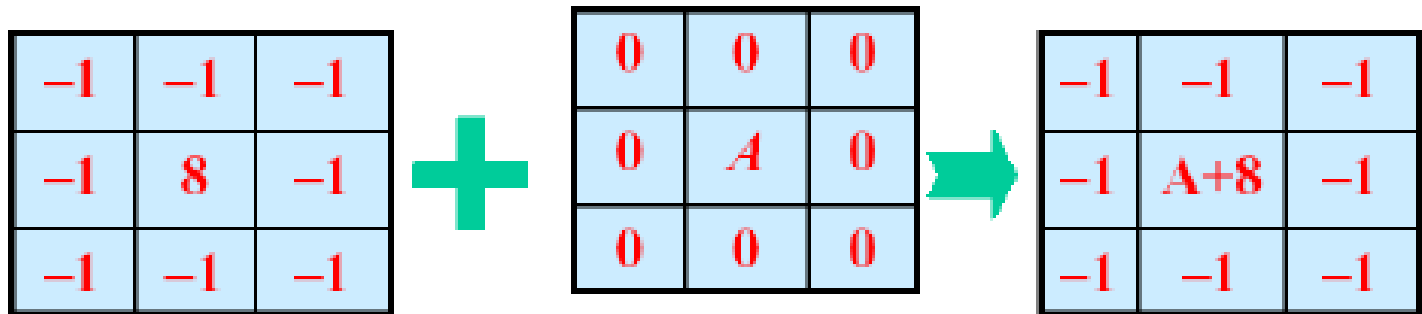
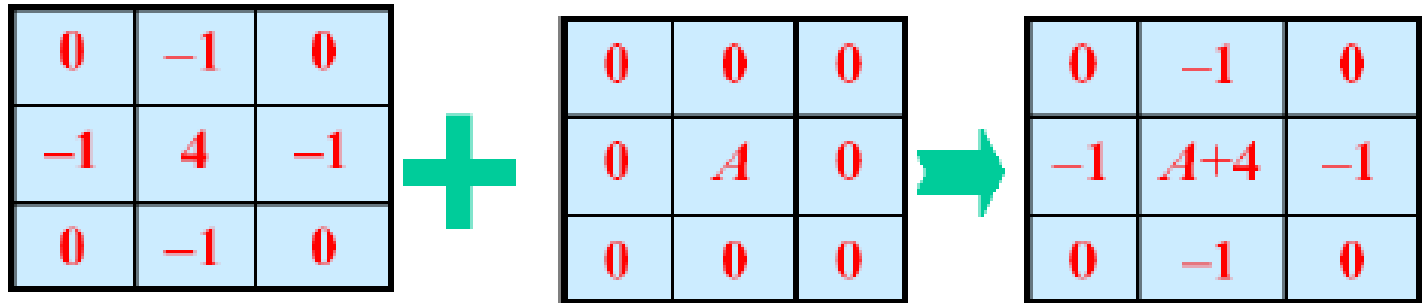
$$\begin{aligned} f_{hb}(x, y) &= Af(x, y) - \bar{f}(x, y), A > 1 \\ &= (A - 1)f(x, y) + f(x, y) - \bar{f}(x, y) \\ &= (A - 1)f(x, y) + f_s(x, y) \\ &\xrightarrow[\text{to } f_s(x, y)]{\text{apply Laplacian mask}} = (A - 1)f(x, y) + [f(x, y) \pm \nabla^2 f(x, y)] \\ &= Af(x, y) \pm \nabla^2 f(x, y) \end{aligned}$$

Principal application:

- High Boost filtering is used when input image is darker than desired
- High-boost filter makes the image lighter and more natural

High Boost Filtering

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$A \geq 1$

High Boost Filtering Masks

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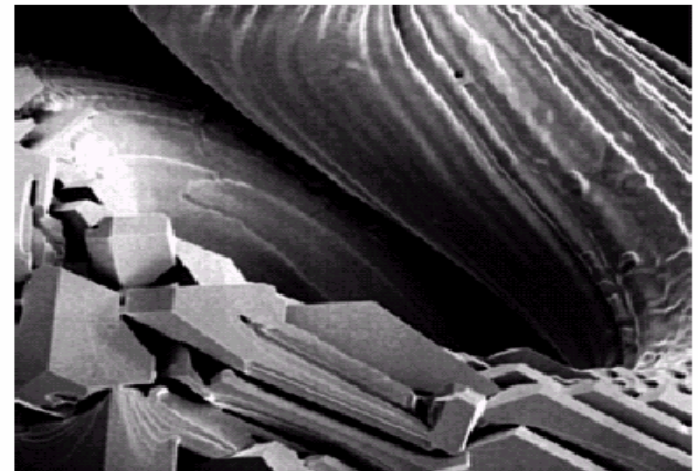
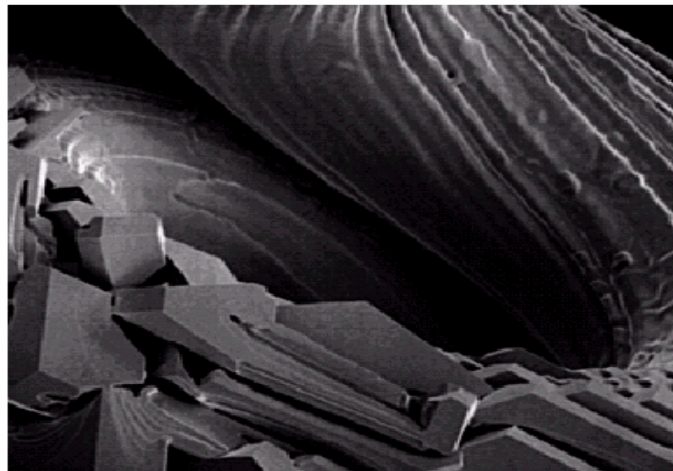
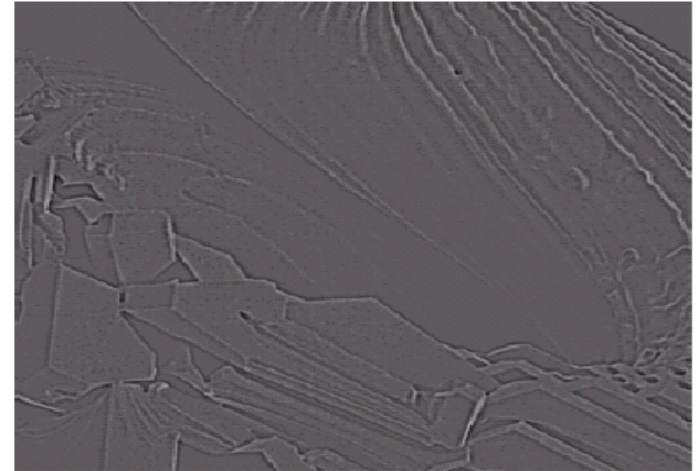
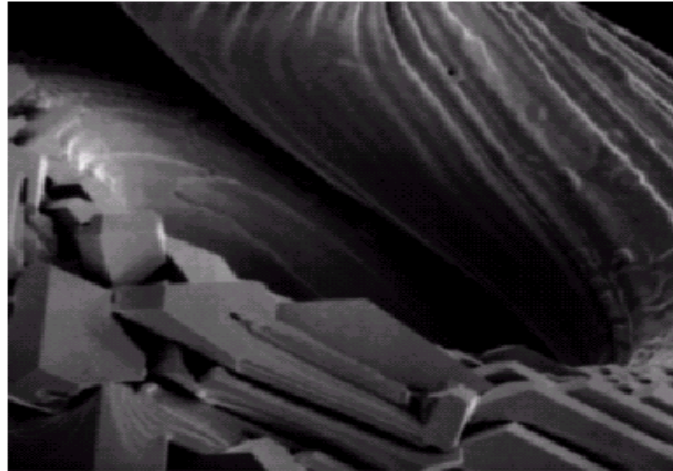
a	b
c	d

FIGURE 3.43

(a) Same as Fig. 3.41(c), but darker.

(b) Laplacian of (a) computed with the mask in Fig. 3.42(b) using $A = 0$.

(c) Laplacian enhanced image using the mask in Fig. 3.42(b) with $A = 1$. (d) Same as (c), but using $A = 1.7$.



High Boost Filtering Masks

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Original Image



0	-1	0	-1	-1	-1
-1	$A + 4$	-1	-1	$A + 8$	-1
0	-1	0	-1	-1	-1

High Boost Filtering Masks

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Mask 1, Sigma = 1



Mask 1, Sigma = 1.5



Mask 1, Sigma = 2



Mask 2, Sigma = 1



Mask 2, Sigma = 1.5



Mask 2, Sigma = 2

References

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1. DIP by Gonzalez

For any query Feel Free to ask



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